

Rebaz Kamal

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Education

University of Massachusetts Boston

Sept. 2026 – Present

Ph.D. in Computer Science

- Research mentors: Dr. Ping Chen and Dr. Wei Ding; Artificial Intelligence & Machine Learning Lab
- Research interests:** robust and trustworthy language models, retrieval-augmented generation, semi-supervised and ensemble learning, AI agent safety and evaluation

University of Massachusetts Boston

Sept. 2022 – May 2026

B.S. in Computer Engineering, Minor in Computer Science

- GPA 3.97/4.0.** Coursework: Applied Machine Learning, Data Structures & Algorithms, Operating Systems, Computer Security, Embedded Systems, Digital Systems, Statistics

Research

RCLA: Relevance-Contrastive Alignment for Robust RAG

Under Review, NAACL 2027

- Developed RCLA, a lightweight training objective that improves language-model robustness to noisy, irrelevant, and counterfactual retrieved evidence without architectural modifications or additional inference-time cost
- Introduced answer-conditioned evidence localization paired with alignment and contrastive losses, separating relevant from defective evidence inside the model's own representations
- Improved Exact Match in **8 of 9** retrieval settings across Qwen2.5-3B-Instruct and Llama-3.2-3B-Instruct

Research Assistant, AI & Machine Learning Lab

May 2025 – Present

University of Massachusetts Boston

- Investigate machine-learning methods spanning ensemble and semi-supervised learning, computer vision, and robust retrieval-augmented language models
- Develop and evaluate learning objectives for robust language-model behavior under noisy and conflicting retrieved evidence; study representation-level signals and failure cases
- Conduct reproducible model training and benchmarking in PyTorch; analyze quantitative results and prepare research visualizations and manuscripts

Experience

Co-Founder & Lead AI/ML Engineer, Airion Technologies

Boston, MA Jan. 2025 – Present

- Co-founded a drone-technology startup developing ML-enabled flight controllers for real-time fault detection and predictive health monitoring
- Lead the development of an AI-driven drone health-monitoring architecture integrating multivariate sensor telemetry, feature extraction, component-level fault classification, and embedded inference for real-time health assessment

Teaching Assistant, University of Massachusetts Boston

Boston, MA Jan. 2024 – Present

- Circuit Analysis I and Lab** (ENGIN 231/271, Sept. 2026 – Present): support 60+ students through lab sessions and coach students through circuit laws, theorems, op-amps, and first- and second-order circuits
- Computer Organization & Assembly** (ENGIN 246, Jan. 2026 – May 2026): guided 40+ students through CPU architecture, datapath design, pipelining, memory hierarchy, and low-level debugging
- Digital Systems with Lab** (ENGIN 241, Jan. – May 2025): helped 30+ students design, simulate, and debug combinational and sequential circuits on FPGAs in VHDL
- Programming in C** (CS 240, Jan. 2024 – Jan. 2026): taught 250+ students pointers, memory management, and data structures, with written code reviews that raised correctness and readability

Technology Intern, Revere Public Schools

Revere, MA Jan.–June 2022

- Resolved hardware and software issues across 500+ staff and student devices and wrote the troubleshooting documentation the district IT team reused

Projects

Citation-First RAG Chatbot, Sustainable Solutions Lab [GitHub](#) [🔗](#)

- Collaborated on Beacon BOT, a retrieval-augmented generation (RAG) assistant designed to help users explore sustainability and climate-related research through evidence-grounded answers with citations.
- Led LLM integration, connecting the language model to the document-retrieval pipeline to generate context-aware, citation-grounded responses from 20 indexed sources (15 PDFs, 5 webpages) comprising 2,354 searchable chunks.
- Contributed to system evaluation, achieving 10/10 citation coverage and evidence support on the official benchmark; expanded testing achieved 49/49 citations, 50/50 evidence support, and 50/50 preferred-source retrieval.

AI Flight Controller for Predictive Health Monitoring [GitHub](#) [🔗](#)

- Collaborated on an AI-enabled drone health-monitoring system designed to identify battery and motor degradation, detect potential component failures, and improve flight reliability through predictive diagnostics.
- Lead all machine-learning development and implementation, including multivariate telemetry feature engineering, fault-classification models, rolling-window analysis, and embedded C++/ArduPilot Lua inference with a weighted 0–100 health score.
- Achieved 93% fault-classification accuracy through simulations and testing with degraded batteries and motors, demonstrating the system's ability to identify component-level health conditions. Presented the drone prototype at the MassRobotics summit

L.R.D.M., AI-Enabled Point-of-Care Diagnostic Device [GitHub](#) [🔗](#)

- Collaborated on a portable AI-powered diagnostic device designed to automate lateral-flow assay interpretation and distinguish disease-associated color patterns without relying on manual visual assessment.
- Served as Machine Learning Lead, directing the development of a ResNet50 CNN using TensorFlow/Keras and transfer learning, with image normalization, augmentation, and stratified training for four assay categories.
- Achieved 92.75% classification accuracy and 0.88 F1-score on held-out test data; collaborated with hardware and software teams to integrate ML inference into a Raspberry Pi-based diagnostic prototype delivering results in under 5 seconds.

Leadership & Outreach

AI Institute Assistant, Paul English Applied AI Institute (PEAII) *Boston, MA Jan. 2026 – Present*

- Co-develop Agentic AI learning modules and hands-on exercises covering LLM tools, API integration, agent workflows, and practical evaluation
- Support the Applied AI Microcredential, including Canvas learning activities, progress assessment, AI demo workshops, and hackathon preparation
- Coordinate AI outreach and technical events connecting students with faculty researchers and industry practitioners

Vice President, UMass Boston Robotics and Engineering Club *Boston, MA Sept. 2025 – Present*

- Lead hands-on workshops in soldering, CAD, and microcontroller programming; support student robotics and drone development
- Participated in the MassRobotics Form & Function Challenge and presented the team's drone prototype at the summit

Skills

Programming Languages: Python, C/C++, Java, SQL, R, MATLAB, Lua

ML Frameworks & Libraries: PyTorch, TensorFlow/Keras, scikit-learn, Pandas, NumPy, Matplotlib

Embedded & Hardware: Arduino, Raspberry Pi, FPGA (VHDL), PCB design, CAD, ArduPilot

Development Tools & Platforms: Git, GitHub, Jupyter, Google Cloud Platform (GCP), VS Code

Awards & Certifications

National Grid Clean Energy Academy Scholarship | Alpha Lambda Delta National Honors Society | Dean's List, every semester

Certifications: Deep Learning and Machine Learning Specializations (DeepLearning.AI); Data Engineering, Big Data, and Machine Learning on GCP (Google Cloud)